Extensive-form games

Nash Equilibrium in Perfect Information Game

$\langle N = \{1, 2\}, S_1, S_2, (u_i)_{i \in N} \rangle$

Theorem (Theorem 2, Kuhn's Theorem)

Every (finite) extensive-form game with perfect information has at least one Nash equilibrium in pure strategies.

Subgame Perfect Equilibrium in Perfect Information Game

Theorem (Theorem 3)

Every (finite) extensive-form game with perfect information has a subgame perfect equilibrium in pure strategies.

- This equilibrium can be obtained using the backward induction principle.
- Theorem ~~1~~ 2 can be seen as a corollary of Theorem ~~2~~ 3.

Idea of the proof of Theorem 2

- We will first discuss the idea of the proof.
- It involves induction on the number of nodes in the game tree.
- Observe that, for a perfect information game, every child of the root of the game tree defines a subgame.

Idea of the proof of Theorem 2

- This means we can view the whole game like this.

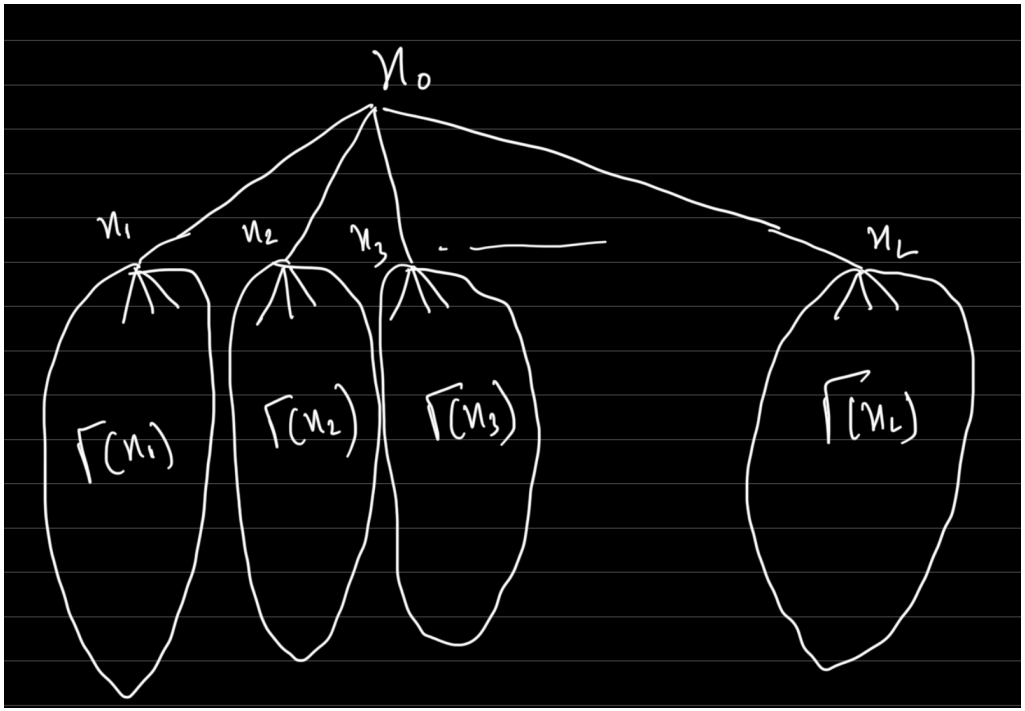

Figure: Figure 1

- Further, each subgame contains fewer nodes than the original game.

Idea of the proof of Theorem 2

- We choose one equilibrium for each such subgame.
- If the root involves a chance move, then the union of the equilibria of all the subgames defines an equilibrium for the entire game.
- If the root is in V_i for some Player i, for each subgame (one for each child), Player i will calculate the utility he will receive under the chosen equilibrium for each subgame, and then choose the node that grants him maximal utility.

Proof of Theorem 2

- We will now formally prove Theorem 2.
- As mentioned before, we will use induction on the number of nodes in the game tree.
- **Base case:** If the game tree has a single node, it means no player has any strategy in the game, and we may say the strategy profile $(\emptyset, \ldots, \emptyset)$ is an NE of the game.

Proof of Theorem 2

- **Induction hypothesis:** Assume that the claim is true for each game in extensive form containing fewer than k nodes.
- We show that it is true for all games with k nodes.
- Let Γ be an extensive form game with k nodes.

Proof of Theorem 2

- Let $x_1, \ldots, x_L$ be the children of the root x_0, and $\Gamma(x_l)$ is subgame whose root is x_l.
- The utility functions of the game Γ are $\hat{u}_i : \times_{i \in N} S_i \to \mathbb{R}$. (Note that this is different from u_i used in the definition of the game).
- For each $l \in \{1, \ldots, L\}$, the utility functions in the subgame $\Gamma(x_l)$ are $\hat{u}_i^l \times_{i \in N} S_i^l \to \mathbb{R}$ where S_i^l is the strategy set of Player i in $\Gamma(x_l)$.

Proof of Theorem 2

- As the root x_0 does not belong to any of the subgames, all these subgames have fewer than k nodes.
- This means each of these subgames has at least one NE in pure strategies (if there are multiple equilibria, we may arbitrarily choose an equilibrium).
- Suppose for the game $\Gamma(x_l)$, the equilibrium is $s^{*l} = (s_i^{*l})_{i \in N}$.
- First, assume that the root is a chance node.

Proof of Theorem 2

- For each $l \in \{1, \ldots, L\}$, let p^l denote the probability that x_l is chosen at the chance node x_0.
- For each Player i, consider the following strategy s_i^*, if the game reaches x_l, implement the strategy s_i^{*l} ~~$s_i{*}l$~~ in the subgame $\Gamma(x_l)$.
- Note that the utility of Player i at this strategy profile $s^* = (s_i^*)_{i \in N}$ is $\hat{u}_i(s^*) = \sum_{l \in \{1,\ldots,L\}} p^l \hat{u}_i^l(s^{*l})$.

Proof of Theorem 2

- We show that s^* is an NE of the game Γ.
- Suppose that Player j deviates to a strategy s_j, and s_j^l is the restriction of s_j to the subgame $\Gamma(x_l)$.
- The utility for Player j under the strategy vector (s_j, s_{-j}^*) is $\hat{u}_j(s_j, s_{-j}^*) = \sum_{l \in \{1,\ldots,L\}} p^l \hat{u}_j^l(s_j^l, s_{-j}^{*l})$.
- Now since s^{*l} is an NE in $\Gamma(x_l)$, we have $\hat{u}_j^l(s_j^l, s_{-j}^{*l}) \leq \hat{u}_j^l(s^{*l})$.
- Therefore, $\hat{u}_j(s_j, s_{-j}^*) = \sum_{l \in \{1,\ldots,L\}} p^l \hat{u}_j^l(s_j^l, s_{-j}^{*l}) \leq \sum_{l \in \{1,\ldots,L\}} p^l \hat{u}_j^l(s^{*l}) = \hat{u}_j(s^*)$.

Proof of Theorem 2

- This means Player j does not have a positive deviation by playing s_j. As both j and s_j are arbitrary, we have s^* is an NE.
- Now assume that the root is a decision node of a Player $i_0 \in N$, i.e., $x_0 \in V_{i_0}$.
- We first define a strategy vector $s^* = (s_i^*)_{i \in N}$, and show that this is an NE.
- For each Player i, $i \neq i_0$, consider the strategy s_i^* as follows: if the game reaches x_l in the first move, choose s_i^{*l} in the subgame $\Gamma(x_l)$.

Proof of Theorem 2

- For Player i_0, define the following strategy $s_{i_0}^{*l}$: at the root x_0, choose a node x_{l_0} such that $l_0 \in \arg\max_{1 \le l \le L} u_{i_0}^{l}(s^{*l})$, i.e., $u_{i_0}^{l_0}(s^{*l_0}) \ge u_{i_0}^{l}(s^{*l})$ for all $l \in \{1, \ldots, L\}$.
- Further, for the subgame starting at x_l, choose the strategy $s_{i_0}^{*l}$.
- The utility under the strategy vector $s^* = (s_i^*)_{i \in N}$ is $\hat{u}^{l_0}(s^{*l_0}) = (\hat{u}_i^{l_0}(s^{*l_0})_{i \in N}$.

Proof of Theorem 2

- We show that s^* is an NE.
- The proof that each Player $i \neq i_0$ cannot get a better utility by deviating to some other strategy is the same as before (for the case when the root is a chance node).
- We will show that Player i_0 also cannot profit from a deviation from $s^*_{i_0}$.
- Suppose Player i_0 deviates by selecting s_{i_0} and let $x_{\bar{l}}$ be the node selected by s_{i_0} at the root. Further for each child x_l, let $s^l_{i_0}$ be the restriction of the strategy s_{i_0} to the subgame $\Gamma(x_l)$.

Proof of Theorem 2

- First assume $\bar{l} = l_0$. Since s^{*l_0} is an NE of the subgame $\Gamma(x_{l_0})$, we have the following

$$\hat{u}_{i_0}(s_{i_0}, s^*_{-i_0}) = \hat{u}^{l_0}_{i_0}(s^{l_0}_{i_0}, s^{*l_0}_{-i_0}) \leq \hat{u}^{l_0}_{i_0}(s^{*l_0}_{i_0}, s^{*l_0}_{-i_0}) = u_{i_0}(s^*).$$

Proof of Theorem 2

- If $\bar{l} \neq l_0$, since s^{*l_0} is an NE of the subgame $\Gamma(x_{\bar{l}})$ and by the definition of l_0, we have the following:

$$\hat{u}_{i_0}(s_{i_0}, s^*_{-i_0}) = \hat{u}^{\bar{l}}_{i_0}(s^{\bar{l}}_{i_0}, s^{*\bar{l}}_{-i_0}) \leq \hat{u}^{\bar{l}}_{i_0}(s^{*\bar{l}}_{i_0}, s^{*\bar{l}}_{-i_0}) \leq \hat{u}^{l_0}_{i_0}(s^{*l_0}_{i_0}, s^{*l_0}_{-i_0}) = u_{i_0}(s^*).$$

- This means Player i_0 cannot benefit by moving to strategy s_{i_0}, completing the proof of the Theorem.

Proof of Theorem 3

- You may try in the same line of arguments. Think about it. (Exercise)